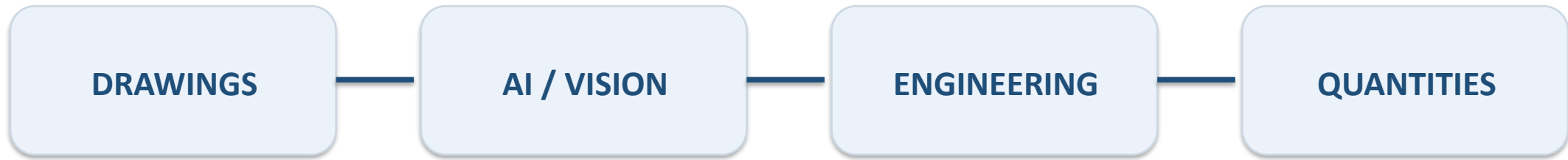


AI-DRIVEN ESTIMATION AUTOMATION



1. Construction Estimation Scope

Major construction materials / work categories identified with the Estimation Team

1. Concrete & Shuttering HIGH PRIORITY Current development Focus Beam & Slab	2. Steel HIGH PRIORITY Current development Focus Beam	3. Blockwork Wall / block quantities
4. Tiles & Granite Floor / wall finishes	5. Plastering Internal / external surfaces	6. Painting Paintable surfaces
7. Waterproofing Wet areas / terraces	8. Doors & Windows Openings / glazing	9. Earthwork Excavation / filling quantities
10. Landscaping Landscape / external works	11. Plumbing & Electrical Plumbing / electrical works	12. MS Grill Work Grill / metal fabrication works

Highlighted focus areas: High Priority - Steel & Concrete

2. Major Structural Components

The structural estimation framework is organized around these core components

1. Beam

Concrete ✓

Shuttering ✓

Steel ✓

Current development focus

2. Slab

Concrete ✓

Shuttering ✓

Steel

Current development focus

3. Column

4. Foundation

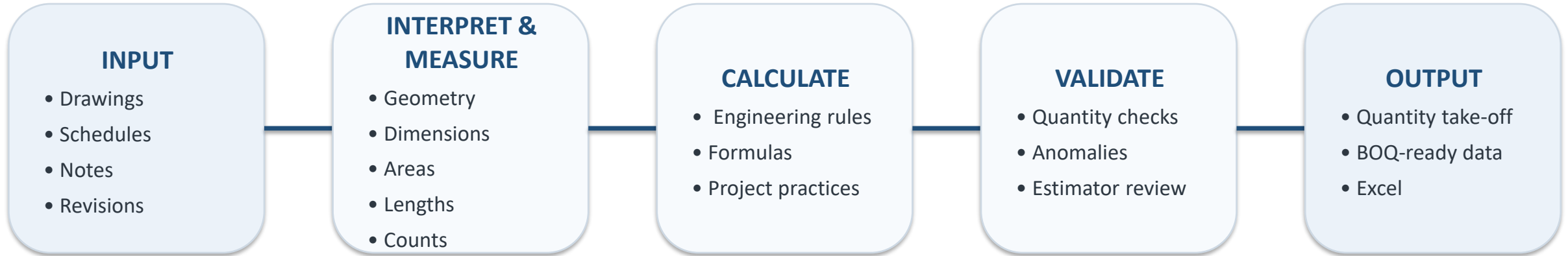
5. Shear Wall

6. Retaining Wall

Highlighted focus areas: High Priority - Beam & Slab

3. What Does Construction Estimation Involve?

From engineering information to measurable, reviewable quantities



AI Model aims to reduce repetitive effort — engineering logic remains central to quantity calculation.

4. Current AI Development Focus

CONCRETE — BEAM & SLAB

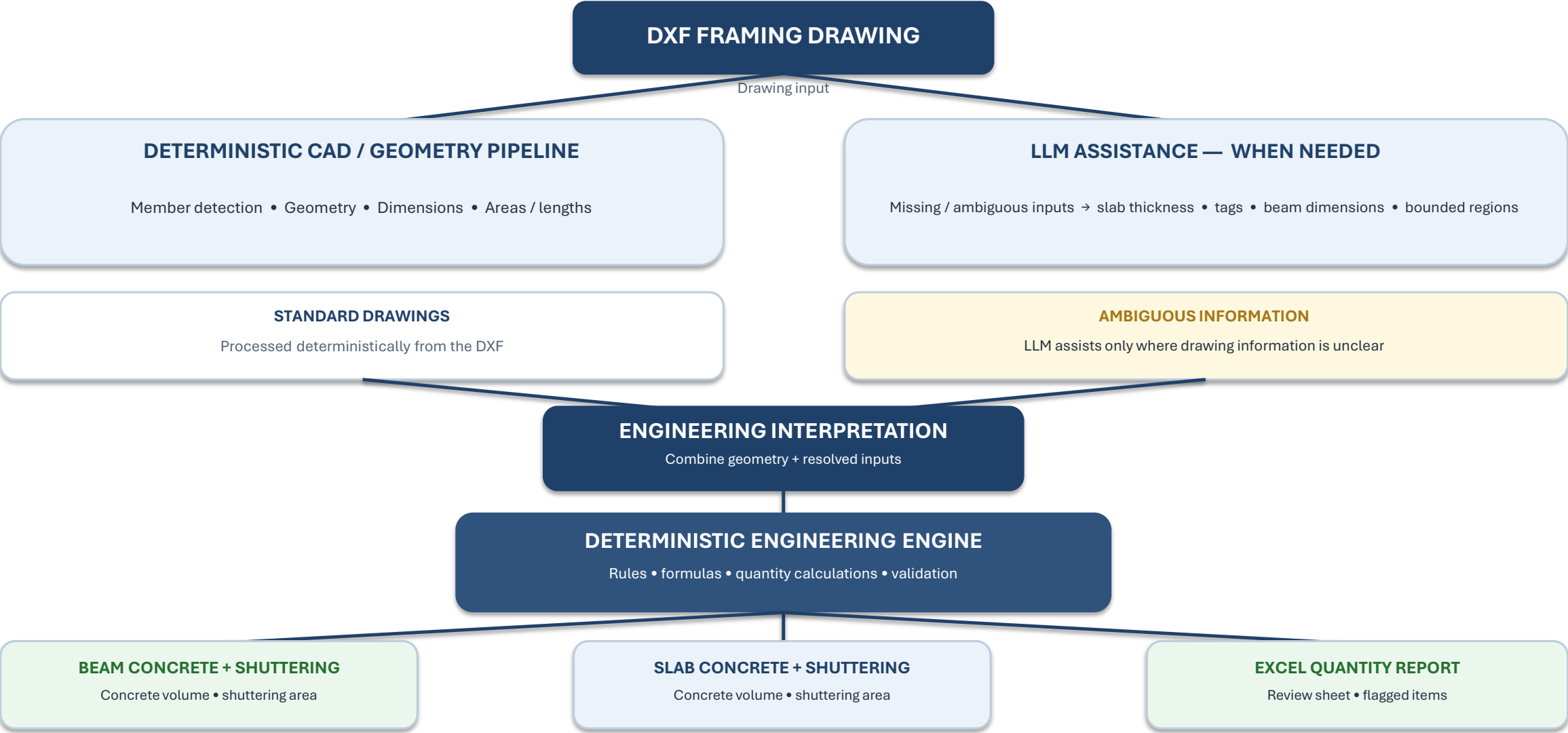
- Beam concrete
- Beam shuttering
- Slab concrete
- Slab shuttering
- Deterministic calculation
- LLM assistance for ambiguous inputs

STEEL — BEAM

- Reinforcement drawing understanding
- Beam matching / ownership
- Reinforcement parsing
- Hybrid CAD + Vision
- Deterministic engineering authority
- Diameter wise Steel Calculations
- Steel quantity + BBS

5. Concrete Beam & Slab — Estimation Architecture

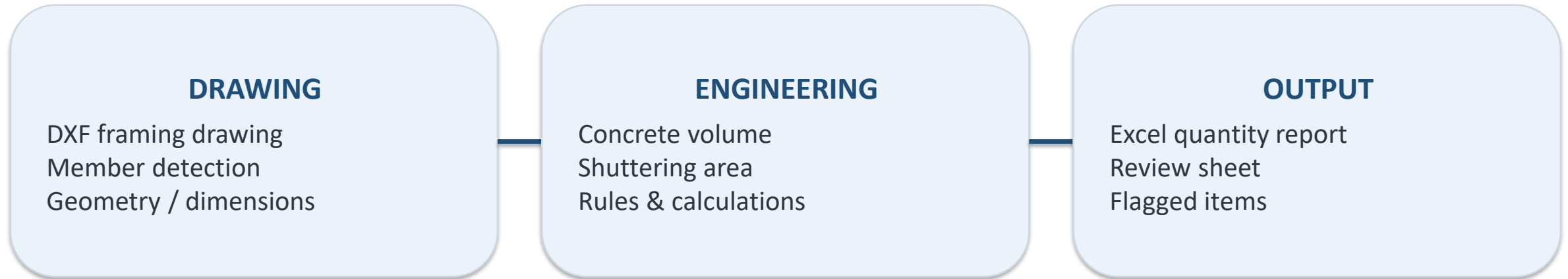
Concrete Beam & Slab Model
<http://13.201.15.89>



Design principle: Standard inputs → deterministic calculation | Ambiguous inputs → LLM assistance → deterministic calculation

5. Concrete Beam & Slab Estimation

Concrete Beam & Slab Estimation Model Link - <http://13.201.15.89>



- Standard drawings can be processed deterministically.
- LLM assistance is used only when information such as slab thickness, tags, beam dimensions or bounded regions is missing / ambiguous.
- Estimation Team validation demonstrated meaningful productivity improvement; quantity accuracy still requires further improvement.

6. Concrete POC - Estimation Team Validation

1

ACCURACY

Current Model Performance

Comparison with manual estimation (Estimation Team)

Work Item	Concrete Accuracy	Shuttering Accuracy
Slab	69%	73%
Beam	78%	81%

2

PRODUCTIVITY

Time Reduction

Reduction in time required for quantity take-off (compared to manual method)

Slab	29% time saving
Beam	67% time saving

3

WHAT THIS MEANS

Key Takeaways

PROCESSING EFFORT REDUCED

Automation delivers clear time savings for quantity take-off.

ACCURACY STILL REQUIRES IMPROVEMENT

Current quantities are not yet accurate enough for direct use.

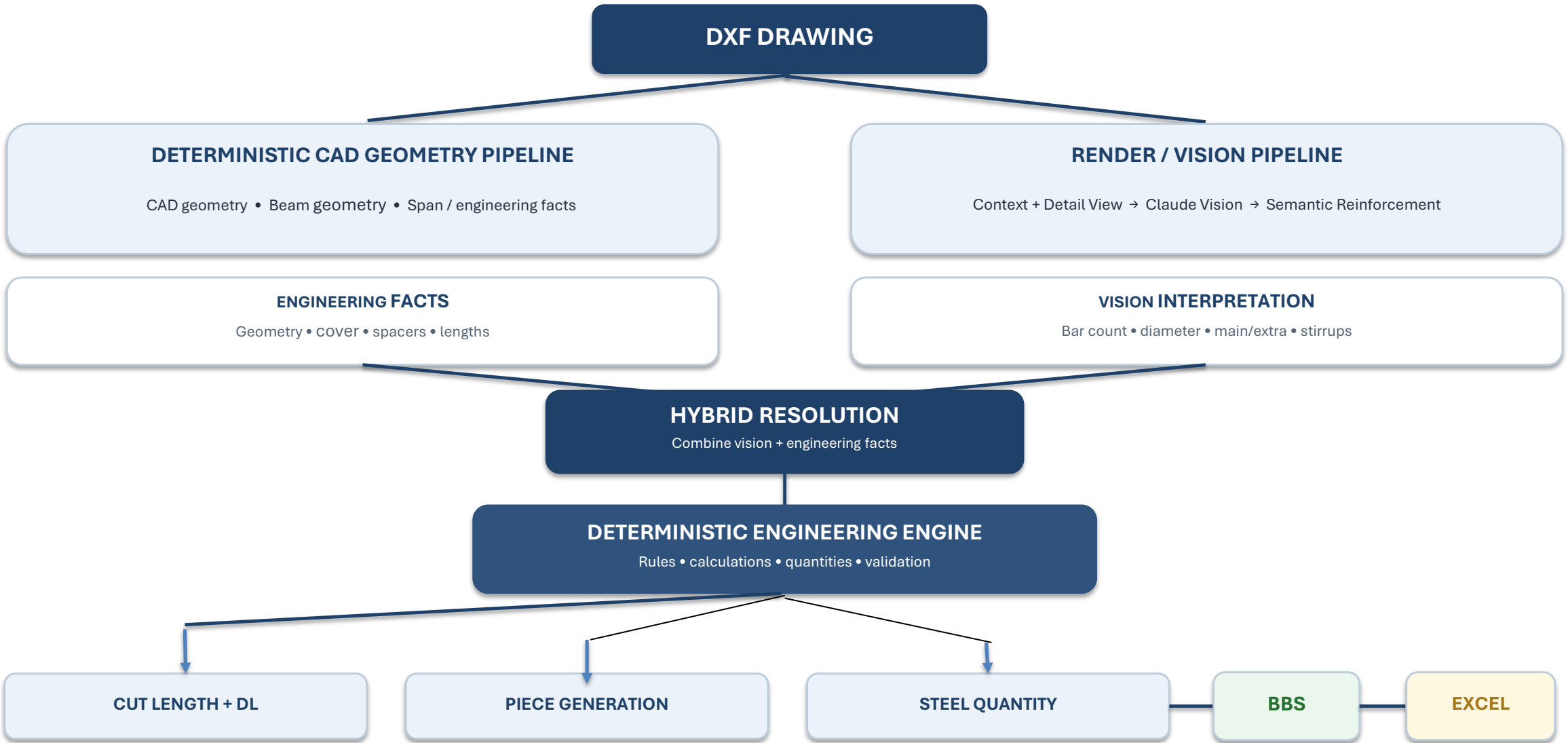
MANUAL VERIFICATION IS CURRENTLY REQUIRED

Results need to be reviewed and corrected by the Estimation Team.

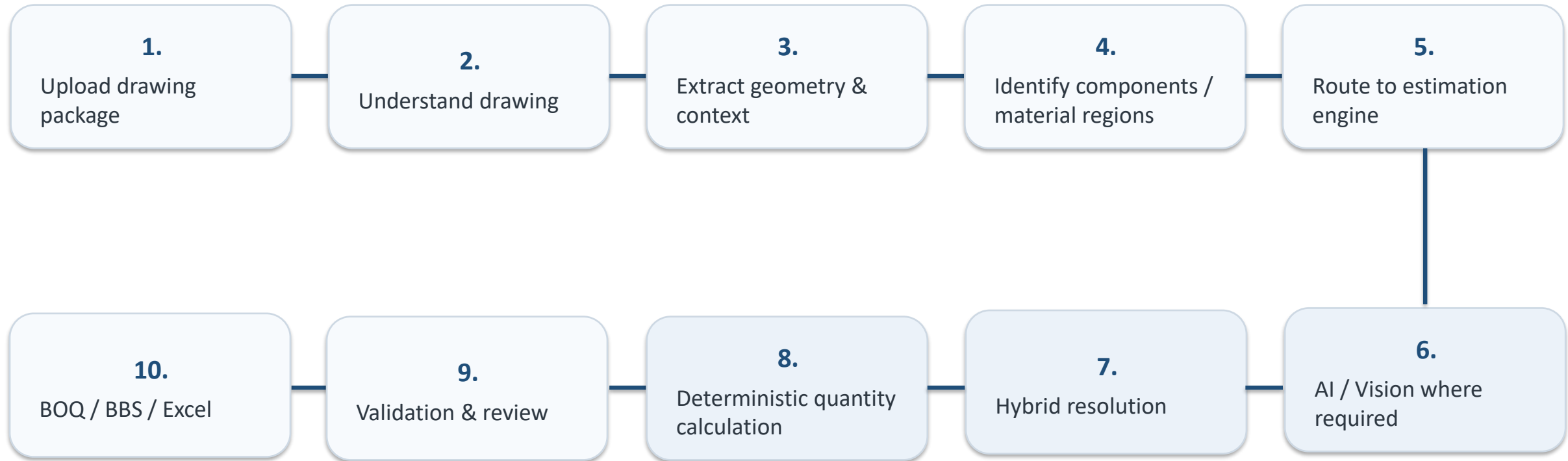
POC CONCLUSION

Automation demonstrated considerable time savings; however, current accuracy requires further development and manual verification before production use.

7. Steel Beam — Hybrid Estimation Architecture



9. End-to-End Steel Estimation Workflow



10. Steel Beam — Internal Accuracy Testing

Current model testing result • Subject to Estimation Team validation

Beam BBS — Automation vs Estimated

Inizio 9th framing

Description	UOM	8	10	12	16	20	25	32	Total
Estimated / Manual	KGS	1,561.1	9,375.2	9,171.8	2,871.9	12,026.0	22,638.4	1,152.0	58,796.3
Automated	KGS	931.1	3,917.6	2,599.0	3,374.8	10,205.7	13,794.8	1,185.3	36,008.4
Difference in quantity	KGS	630.0	5,457.6	6,572.7	502.9	1,820.3	8,843.5	33.4	22,787.9
Error Scale		Underestimate	Underestimate	Underestimate	Overestimate	Underestimate	Underestimate	Overestimate	Underestimate
% Accuracy	-	60%	42%	28%	82%	85%	61%	97%	61%

Key Observations from Estimation Team

- 01

BAR DIAMETER

Top bars specified as 16 mm in the drawing are sometimes read as 20 mm in the output.
- 02

STIRRUPS

Number of stirrups calculation is incorrect in some cases.
- 03

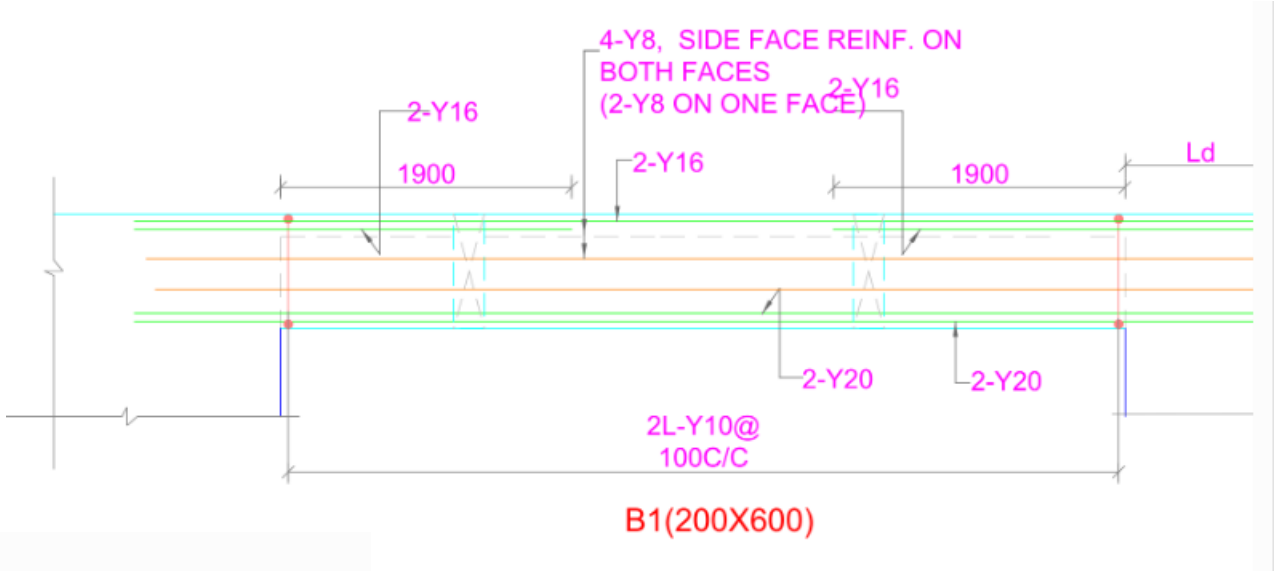
SPACER BARS

Spacer diameter and quantity are not consistently taken as per General Notes; same beam may be counted twice.
- 04

BEAM SIZE

Beam length is not always as per the framing plan, causing slight underestimation.
- AI Construction Estimation | Steel Beam

Cropped Beam Image Sent to Claude



Claude/ Hybrid Answer

B1 — B1(200×600)

B1 detail crop

Claude / Hybrid answer:

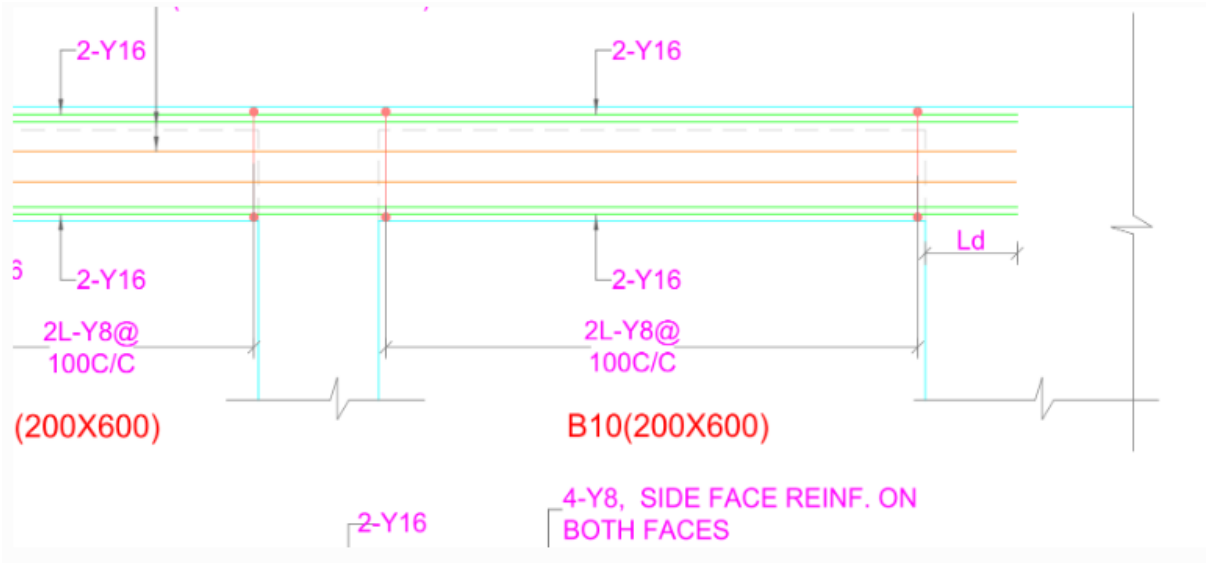
Group	Layer	Spec	Role	Scope
G1	TOP	2-Y16	MAIN	FULL_SPAN
G2	TOP	2-Y16	EXTRA	LEFT_SUPPORT
G3	TOP	2-Y16	EXTRA	RIGHT_SUPPORT
G4	BOTTOM	2-Y20	MAIN	FULL_SPAN
G5	BOTTOM	2-Y20	EXTRA	RIGHT_SUPPORT
G6	SIDE_FACE	4-Y8	EXTRA	FULL_SPAN

Stirrups (Vision): 2L-Y10@100C/C

Estimator Interpretation

SI no		Description	No./Dia.	B(m)/ No.
		SHR & OHT Top		
1.00	TF	B1	1.00	0.20
	TF	Top bars	16.00	2.00
	TF	Top bars - Extra	16.00	2.00
	TF	Bottom bars	20.00	2.00
	TF	Bottom bars	20.00	2.00
	TF	Stirrups	8.00	57.00
	TF	SFR	8.00	4.00

Cropped Beam Image Sent to Claude



Claude/ Hybrid Answer

B10 — B10(200×600)

B10 detail crop

Claude / Hybrid answer:

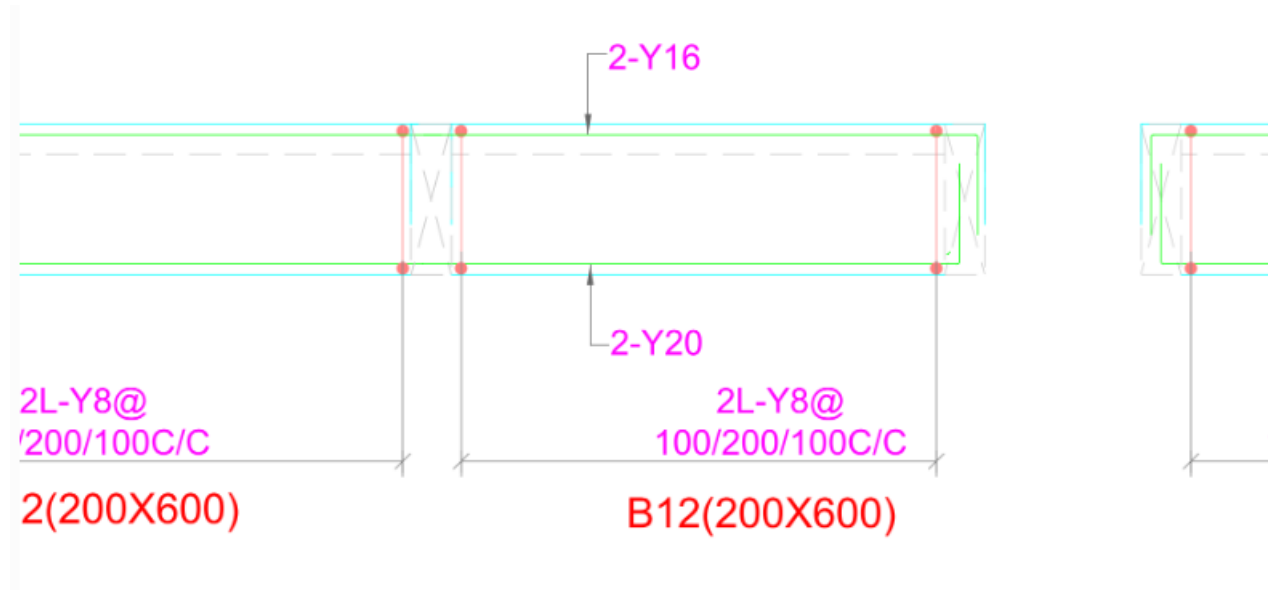
Group	Layer	Spec	Role	Scope
G1	TOP	2-Y16	MAIN	BOTH_SUPPORTS
G2	TOP	2-Y16	EXTRA	LEFT_SUPPORT
G3	TOP	2-Y16	EXTRA	RIGHT_SUPPORT
G4	BOTTOM	2-Y16	MAIN	FULL_SPAN
G5	SIDE_FACE	4-Y8	—	—

Stirrups (Vision): 2L-Y8@100C/C

Estimator Interpretation

10.00	TF	B10	1.00	0.20
	TF	Top bars	16.00	2.00
	TF	Top bars	16.00	2.00
	TF	Bottom bars	16.00	2.00
	TF	Bottom bars	16.00	2.00
	TF	Stirrups	8.00	40.00
	TF	SFR	8.00	4.00

Cropped Beam Image Sent to Claude



Claude/ Hybrid Answer

B12 — B12(200×600)

 B12 detail crop

Claude / Hybrid answer:

Group	Layer	Spec	Role	Scope
G1	TOP	2-Y16	MAIN	FULL_SPAN
G2	BOTTOM	2-Y20	MAIN	FULL_SPAN

Stirrups (Vision): 2L-Y8@100/200/100C/C

Estimator Interpretation

12.00	TF	B12	1.00	0.20
	TF	Top bars	16.00	2.00
	TF	Bottom bars	20.00	2.00
	TF	Stirrups	8.00	45.00
	TF	Stirrups	8.00	11.00

Why Estimation Automation Takes Significant Time

Development effort varies by work item • Engineering understanding + diagnostics + validation

1. Engineering Complexity

Different drawings, structural rules, quantity logic and calculation methods for each work item.

2. Drawing Understanding

Understand how information is represented, connected and interpreted in real construction drawings. This must be transferred to the System.

3. Root-Cause & Diagnostics

Trace incorrect results through multiple stages to identify the actual source of the problem.

4. Solution & Iteration

Implement possible solutions and test them across different drawing conditions and edge cases.

5. Validation & Reliability

Validate outputs against engineering expectations before the automation can be trusted.

Development time = Understanding → Diagnosis → Solution → Iteration → Validation